FARM MANAGEMENT SYSTEM

USER STORY:

The Form Management System is designed to provide a simple and organized platform for handling digital forms and their submitted information. The system allows administrators and authorized users to create forms, maintain form details, and manage submitted records from a single Salesforce-based application.The system enables users to enter the required information through structured forms and ensures that important fields are properly validated before the data is saved. This helps reduce incomplete or incorrect entries and makes the overall form-handling process more reliable.Administrators can monitor submitted forms, update records when necessary, and use reports and dashboards to understand the available information. The system also provides controlled access to form data through appropriate user roles and permissions.By moving the form management process from manual methods to a centralized digital platform, the system helps reduce paperwork, improve record organization, simplify data management, and support faster access to important information.

1. INTRODUCTION

1.1 Project Overview:

The **Form Management System (FMS)** is a Salesforce-based application developed to manage forms and their related records in an organized digital environment.The system provides a common platform for creating forms, entering information, maintaining submitted records, and accessing stored data. It brings different form-related activities into one application instead of handling them separately.Through Salesforce features such as custom objects, fields, relationships, validation, user access, reports, and dashboards, the system provides a structured method for managing form information.

1.2 Purpose:

The main purpose of this project is to:

- Digitize agricultural data management.
- Improve data accuracy and accessibility.
- Automate record handling.
- Enable real-time monitoring.
- Support data-driven farming decisions.
- Enhance productivity and transparency.

2. IDEATION PHASE

2.1 Problem Statement

ID	Persona	Goal	Problem	Current State	Emotion
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	Frustrated
PS-2	Farm Manager	Monitor productivity and sales	Reports are delayed and inaccurate	Data is scattered across files	Stressed
PS-3	Buyer	Access availability and pricing	Information is not updated regularly	No centralized digital system	Uncertain
PS-4	System Admin	Manage access and security	Unauthorized access may occur	Access control is handled manually	Concerned

2.2 Empathy Map Canvas

Target Users: Farmers, Farm Managers, Buyers, System Administrators.

Empathy Analysis:

- **Think & Feel:** Need for accurate and timely farm data to ensure profitability.
- **Hear:** Feedback from agricultural officers and market trend reports.
- **See:** Physical registers, overflowing paper files, and fragmented spreadsheets.
- **Say & Do:** Entering records manually, following traditional seasonal cycles.
- **Pain:** Risks of data loss, manual entry errors, and significant reporting delays.
- **Gain:** Process automation, instantaneous reporting, and cloud-based security.

2.3 Brainstorming:

Selected Idea: **Salesforce-based Farm Management System.**
The team evaluated manual records, Excel-based management, and custom web/mobile apps, but chose Salesforce for its robust cloud infrastructure, security, and built-in analytics capabilities.

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map:

1. **Login:** User authenticates via the Salesforce portal.
2. **Access App:** Navigation to the Farm Management System Lightning App.
3. **Data Entry:** Recording farm and crop data.
4. **Registration:** Onboarding farmers and buyers.
5. **Reporting:** Running yield and transaction reports.
6. **Analysis:** Reviewing interactive dashboards.

7. **Decision Making:** Strategic planning based on real-time data.

3.2 Solution Requirements

Functional Requirements:

- **FR-1:** Secure User Authentication.
- **FR-2:** Management of Farm, Crop, Farmer, and Buyer records.
- **FR-3:** Data Entry with strict Validation Rules.
- **FR-4:** Complex Relationship Mapping between objects.
- **FR-5:** Custom Report Generation.
- **FR-6:** Multi-component Dashboard Visualization.
- **FR-7:** Granular User Role Management.

Non-Functional Requirements

- **Usability:** Simple Lightning UI for non-technical users.
- **Security:** Profiles and roles to ensure data privacy.
- **Reliability:** Automatic cloud backups.
- **Performance:** Real-time reporting with response times under 3 seconds.
- **Availability:** 24/7 cloud-based access.

3.3 Data Flow Diagram:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards

3.4 Technology Stack:

- **Frontend:** Salesforce Lightning UI
- **Backend:** Salesforce Platform
- **Database:** Salesforce Cloud Database
- **Automation:** Validation Rules & Flow

- **Analytics:** Salesforce Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem-Solution Fit:

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely agricultural insights for better crop rotation and sales strategies.

4.2 Proposed Solution:

- **Problem:** Manual maintenance leading to errors and delays.
- **Solution:** Salesforce-based centralized management.
- **Innovation:** Automated validation and cloud-native mobile-ready reporting.
- **Impact:** Reduced manual overhead and high data integrity.
- **Scalability:** Supports multi-region farm growth.

4.3 Solution Architecture:

1. **Presentation Layer:** Salesforce Lightning Experience.
2. **Application Layer:** Logic, Validation Rules, and Object Relationships.
3. **Data Layer:** Salesforce Multi-tenant Cloud Database.
4. **Analytics Layer:** Native Report Engine and Dashboard Builder.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning:

The project follows an Agile approach, divided into sprints focused on setup, data modeling, automation, and security.

Product Backlog & Sprint Schedule

Sprint	Task	User Story	Points	Priority
Sprint-1	Dev Setup	Configure Salesforce developer account and environment.	3	High
Sprint-2	Data Modeling	Create Farm, Crop, Farmer, and Buyer objects and relationships.	5	High
Sprint-3	Automation	Implement validation rules and automated status updates.	3	High
Sprint-4	Security	Define Profiles, Roles, and Sharing Rules.	5	High
Sprint-5	Reporting	Design yield, sales, and transaction reports.	4	High

Project Tracker

Sprint	Points	Duration	Start Date	End Date	Status
Sprint-1	20	6 Days	Date	Date	Completed
Sprint-2	20	6 Days	Date	Date	In Progress

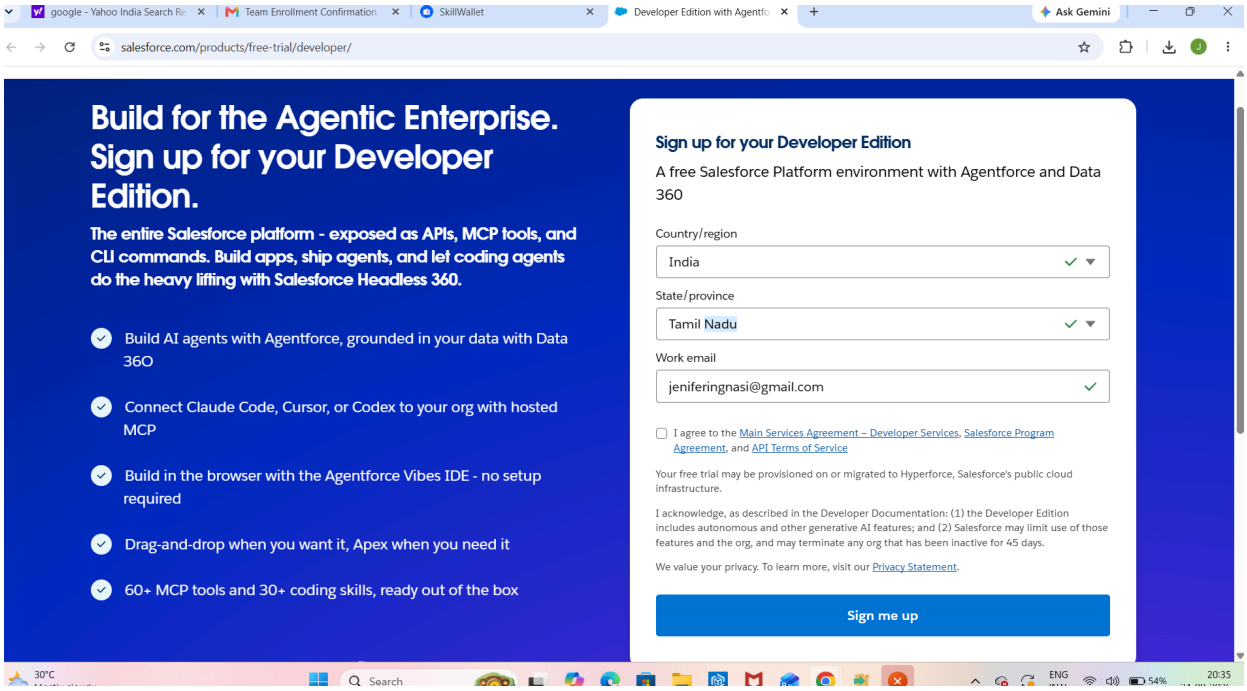
6. PROJECT DEVELOPMENT PHASE

Milestone 1: Salesforce Environment

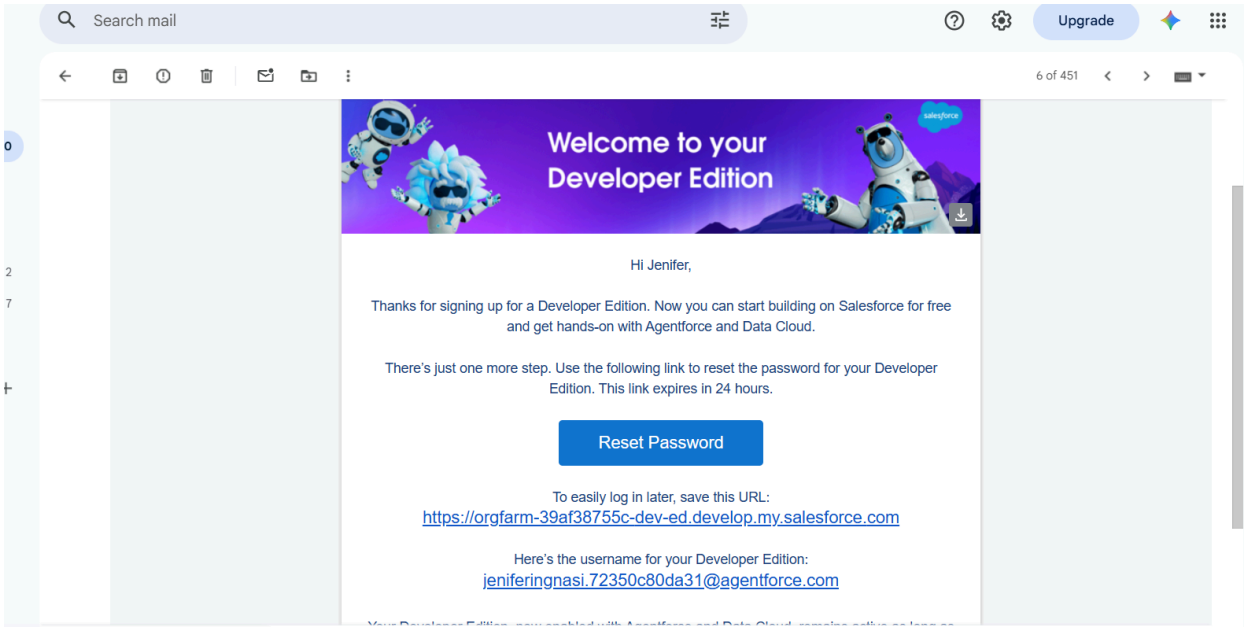
The development environment was initiated using a Salesforce Developer Org. This provides a sandbox-like environment for building custom logic and objects.

Activity1:

Creating a Salesforce Developer Org



Activity 2:Account Activation



Milestone 2: Objects

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an
organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

1.Farm

2.Crop

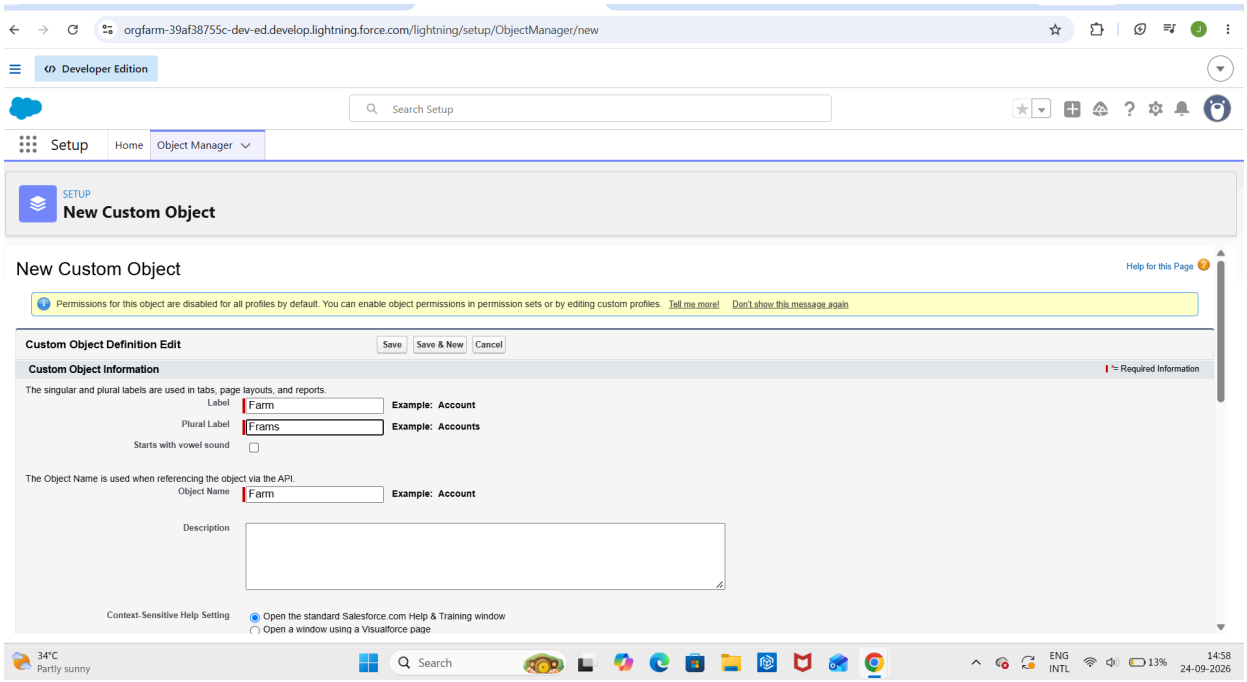
3.Farmer

4.Buyer

Activity 1:

Farm: Primary asset record.

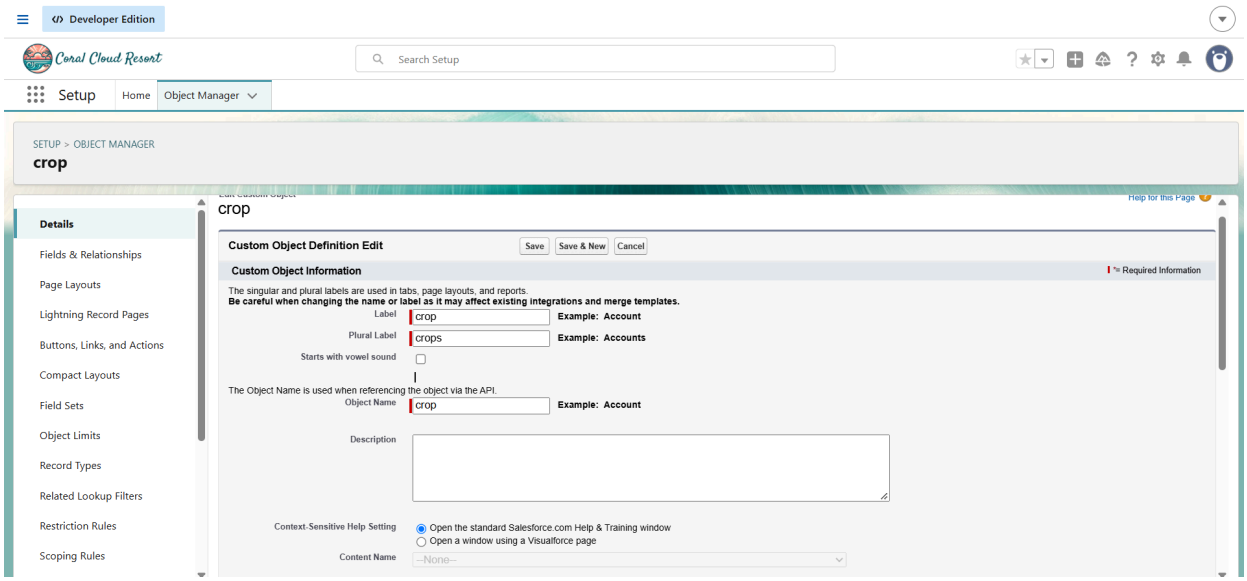
- From setup click on object manager.
- Click create, select custom object.



- Fill in the label as " Farm ".
- Fill in the plural label as " Farms".
- Record name: Farm Name"
- Select the data type as "Text".
- Leave everything else as is, and click Save.

Activity 2:
Crop: Tracks individual planting cycles.

- Click on the gear icon and then select Setup.
- Click on the object manager tab just beside the home tab.
- After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object. a. On the Custom Object Definition page, create the object as follows:



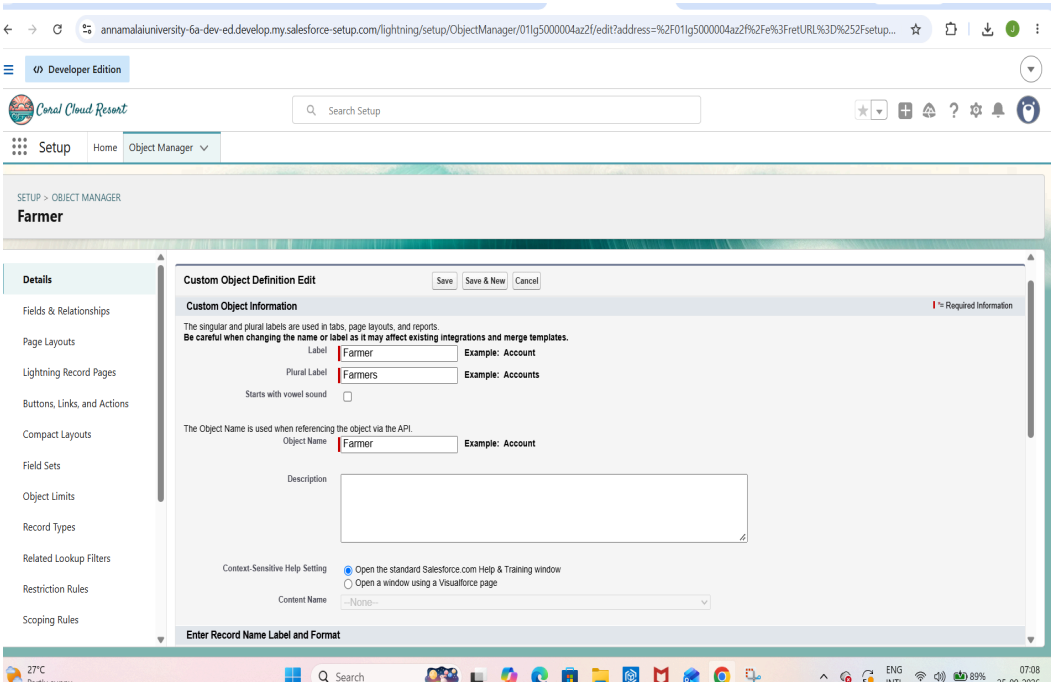
- Label: Crop
- Plural Label: Crops
- Record Name: Crop Name
- Select the data type as "Text".
- Check the Allow Reports checkbox
- Check the Allow Search checkbox
- In the Object Creation Options section, select Add Notes and Attachments related list to default page layout
- Click Save.

Activity 3:

Farmer: Profiles for the agricultural workforce.

- Click on the gear icon and then select Setup.
- Click on the object manager tab just beside the home tab.

  1.After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object

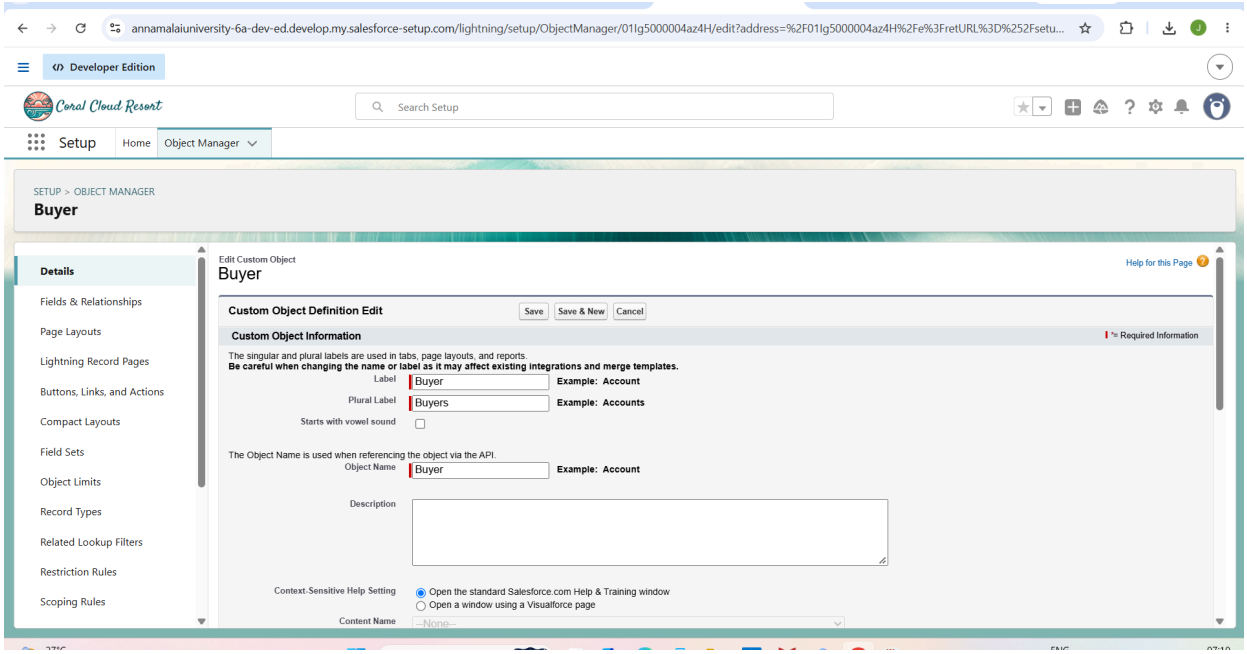


- On the Custom Object Definition page, create the object as follows:
- Label: Farmer
- Plural Label: Farmers
- Record Name: Farmer Nameb
  Select the data type as "Text".
- f. Check the Allow Reports checkbox.
- g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
- i. Click Save.

Activity 4:

Buyer: Records for commercial entities and individuals.

- Click on the gear icon and then select Setup.
- Click on the object manager tab just beside the home tab.
- After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object. a. On the Custom Object Definition page, create the object as follows:



- Label: Buyer
- Plural Label: Buyers
- Record Name: Buyer Name
- Select the data type as "Text".
- Check the Allow Reports checkbox.
- Check the Allow Search checkbox.
- In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.``
- Click Save.

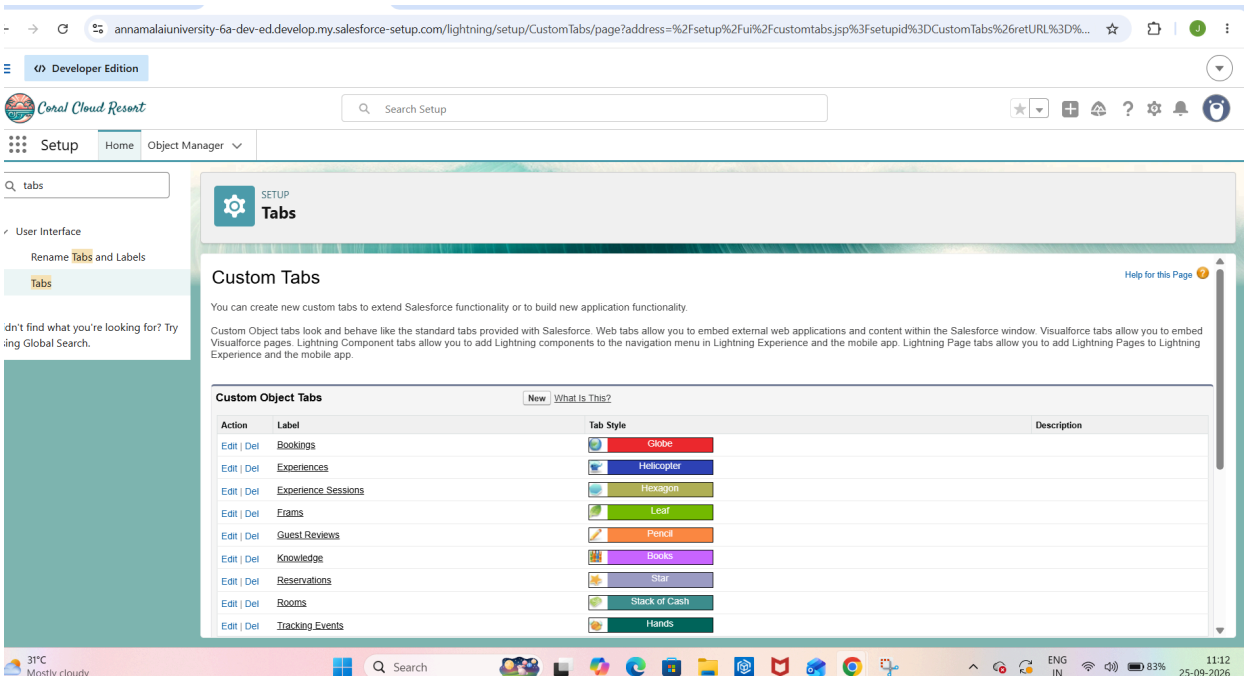
Milestone 3 : Tabs

Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

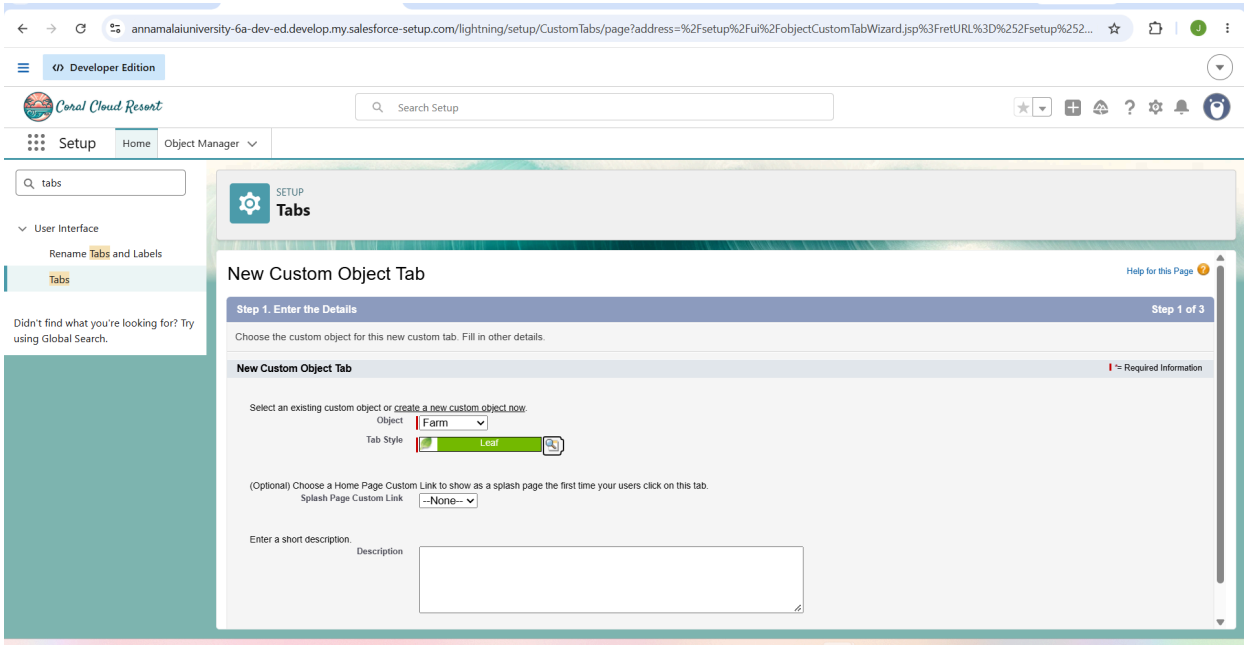
Activity 1:
Creation of Farm tab

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.



2. Under custom object tabs, click New.



3. For Object, select Farm.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click NextNext, and Save.

Activity 2:

Creation of Crop tab

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, select any icon.

5. Leave all defaults as is. Click NeCreation of Buyer tab Next, Next, and Save.

Activity 4:

Creation of Buyer tab

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Milestone 4: Lightning App

What is an App?

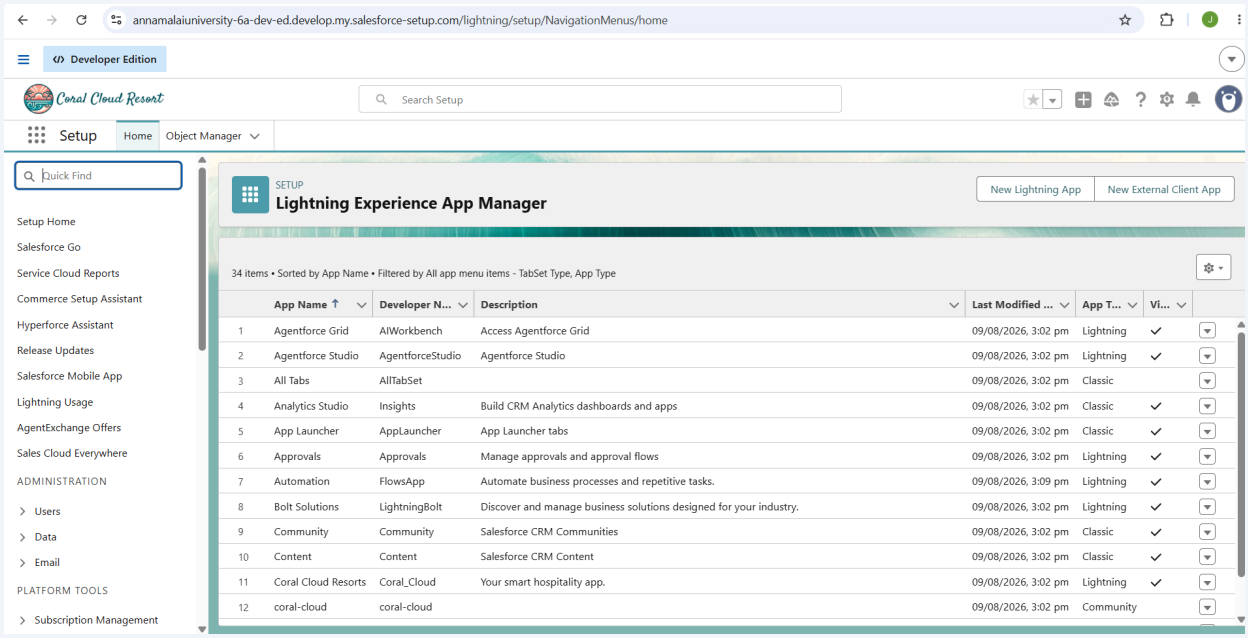
Apps in Salesforce area group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:

Create the Farm Management System app

- From Setup, enter App Manager in the Quick Find and select App Manager.

- Click New Lightning App. Farm Management System as the App Name, then click Next



Under App Options, leave the default selections and click Next.

● Under Utility Items, leave as is and click Next.

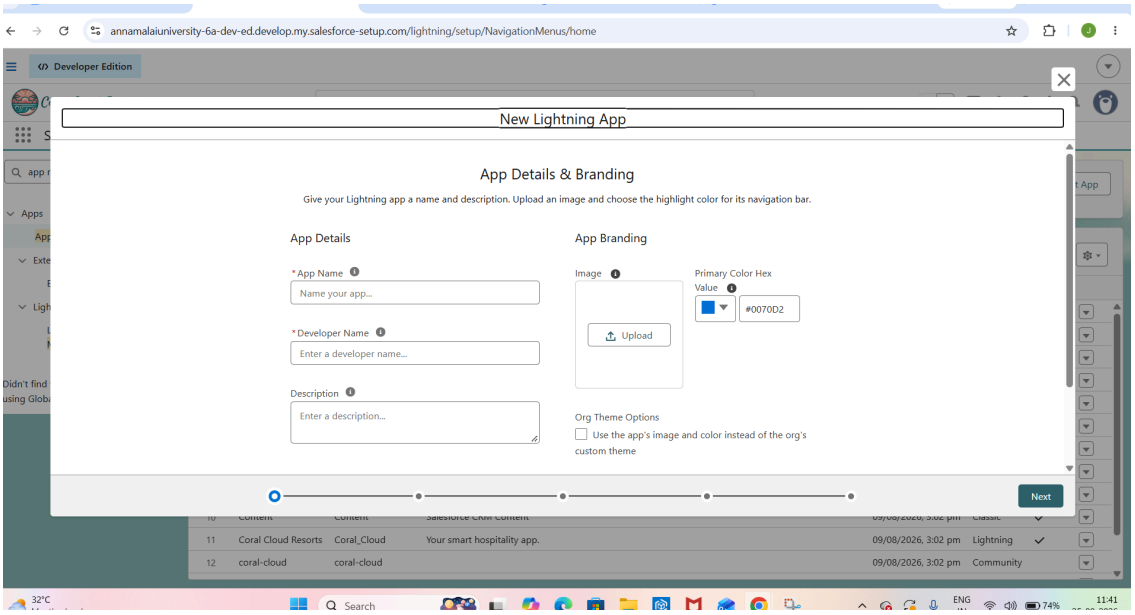
● From Available Items, select Farm, Crop, Farmer ,Buyer Reports ,and Dashboards and move them to Selected Items. Click Next.

● From Available Profiles, select System Administrator and move it to Selected Profiles. Click Save & Finish.

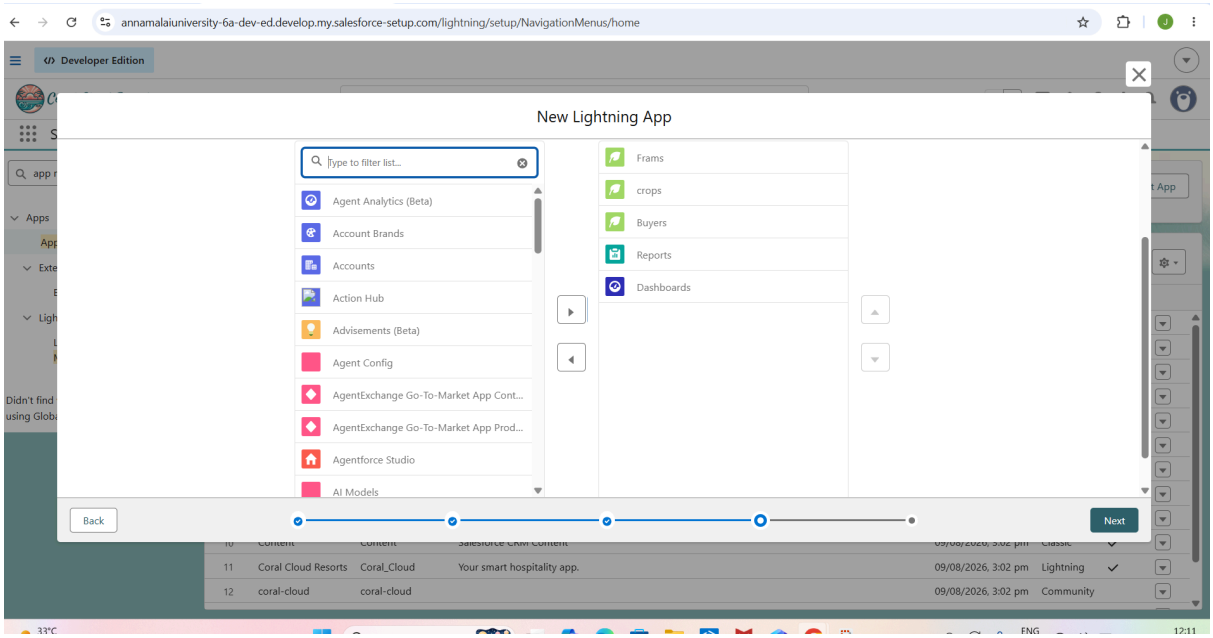
● To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.

Note:

1. App Launcher-Displays available apps.
2. App Name-Displays the current selected app.



3. Navigation menu-Displays the tabs available inside the app.



Milestone 5: Fields and Relationships

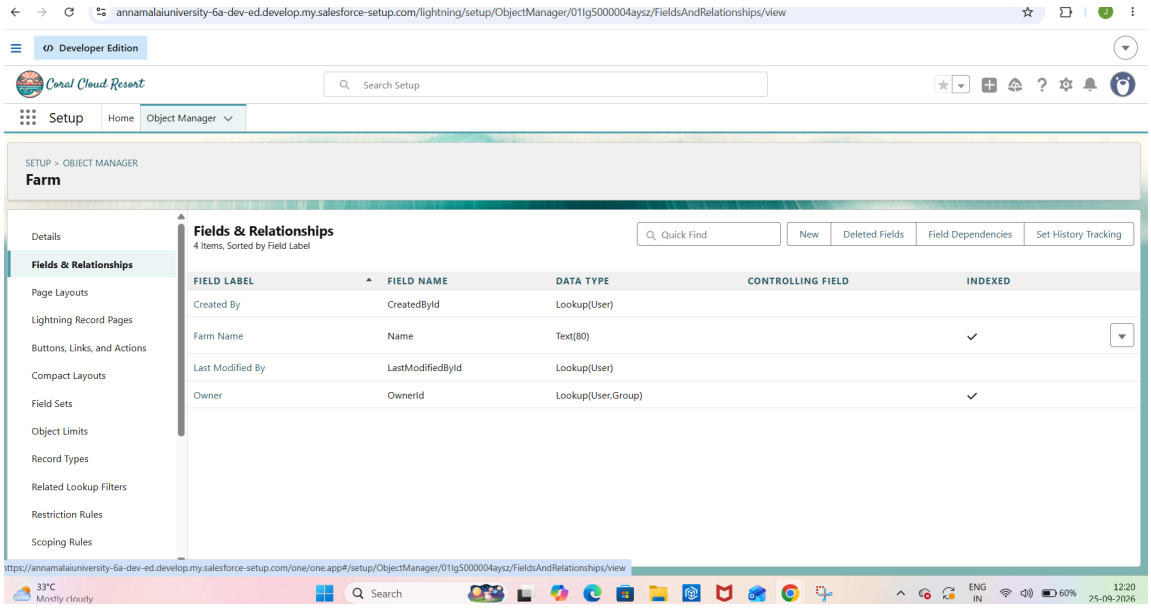
What are fields?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

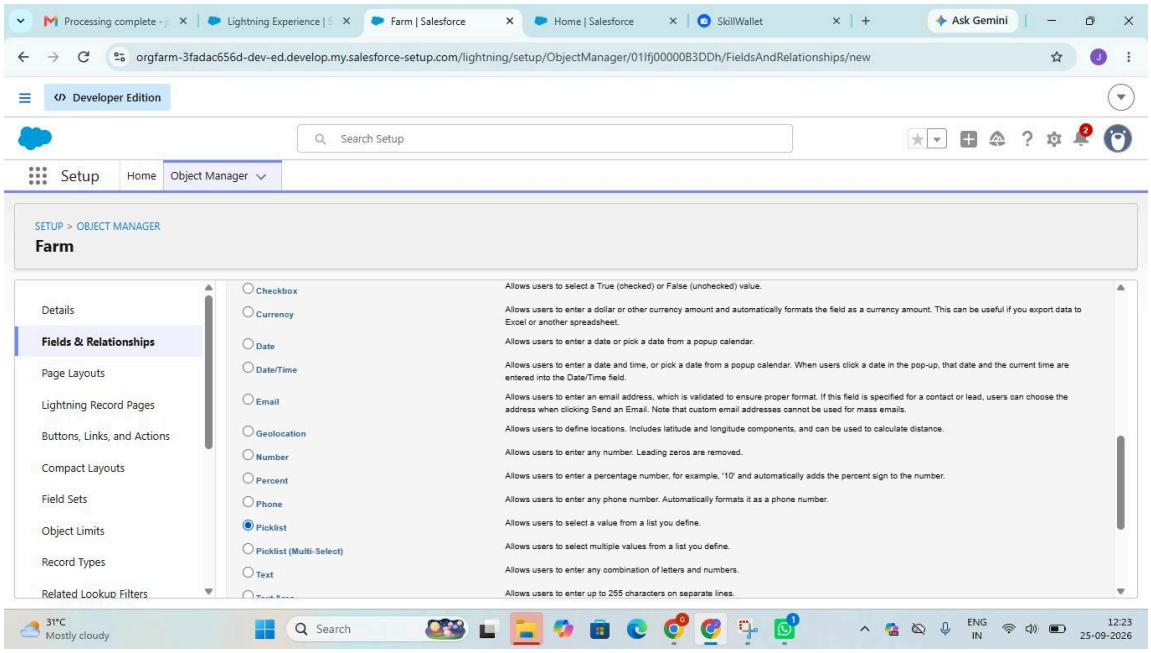
Activity 1:
Creation of fields for the Farm objects

- **Farm:** Picklists for Soil Quality and Irrigation; Master-Detail to Farmer

1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New

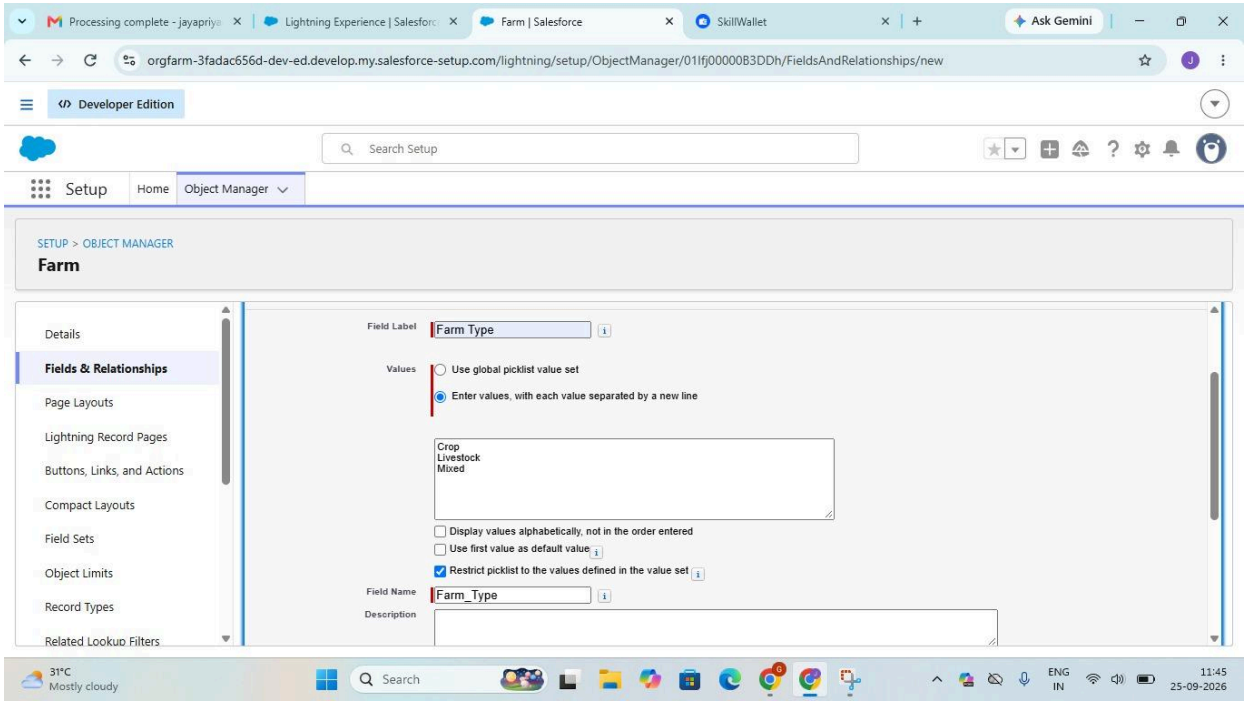


5. Choose Data type as a picklist ,Click next



6.Enter Field Label (Farm Type),and Enter Values as (Crop,Livestock,Mixed)

7.Next,Next Click Save.



Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

1.Next we Create another Picklist Field,Enter field label (Soil Quality) enter values (Organic,Cover Crops,Composting)

2.Again we Create Picklist field,Field Label (Irrigation type),

Enter Values (Sprinkler irrigation,Drip irrigation,localized irrigation)

3.Choose data type as a text,enter field label(Location)

4.Choose data type as text,enter field label (size)

5.Create a Master Detail relationShip with Farmer Object.

Activity 2:

Creation of fields for the Crop objects

- **Crop:** Planting/Harvest dates; Lookup to Farm; Formula fields for Yield calculation.

1.Choose Data type As a text,Enter Field Label name (Crop Name),Next save.

2. Select Data type as date ,Enter Field label (Planting Date),then save .

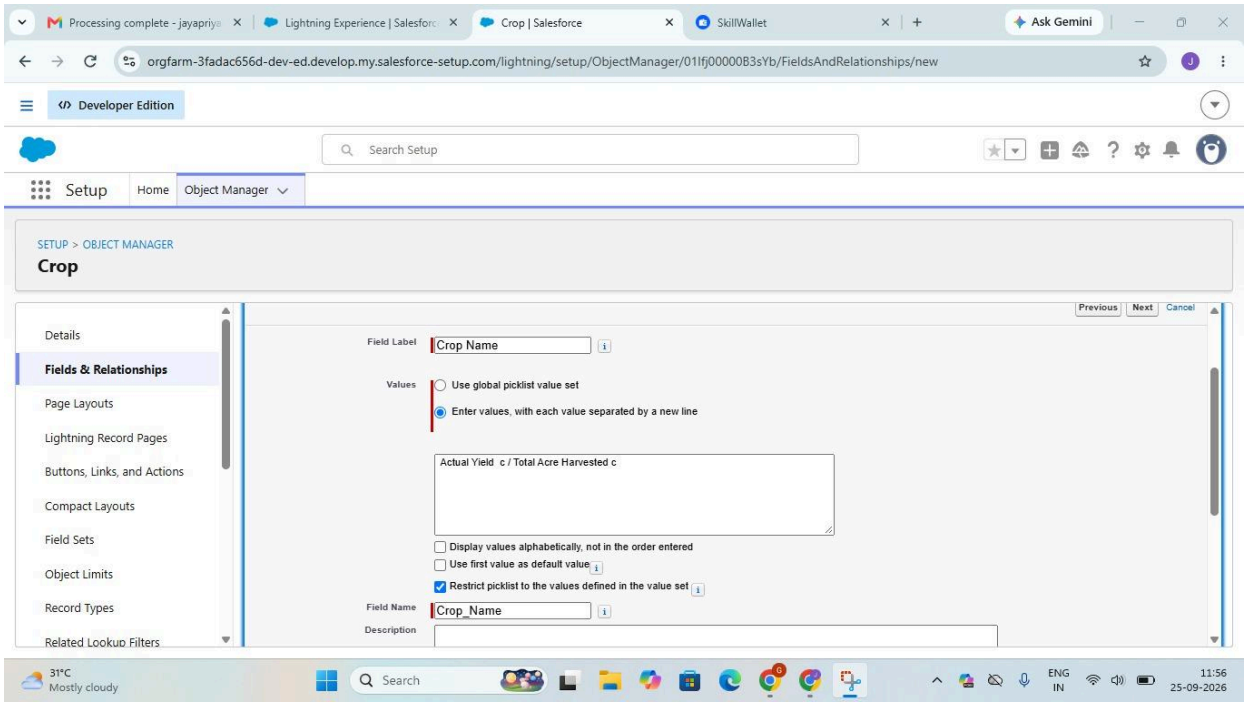
3.Again we select data type as a date,Enter field label (Harvest date),save

4.Next We Create Lookup relationships on crop objects related to Farm Objects.

5.We select data type as currency,Enter field Label (Actual Yield), Next Save it.

6.Select Data type as a Number,enter Field Label (Total Acres Harvested),Enter Maximum Length(20) and save.

7.Now We create a Formula Field For That Select data Type As a Formula,Click Next.



Enter Field Label (Expected Yield per Acre),Select formula return type as Currency,click Next.

9.Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.

Activity 3:

Creation of fields for the Farmer objects

- **Farmer:** Phone and Address details; Field dependencies for Location (State/District).

1.Select Data type Phone,Enter field label(Contact Number),next Save.

2.Select Data type Picklist,Enter Field Label(State),Enter values (Telangana,Andhra Pradesh,Tamilnadu),save.

3.Choose Data type Picklist,Enter Field Label (District),Enter Values of district,next save.

4.Again select Data type as Picklist,Enter Field Label (Village),Enter Values of villages,next save.

Creation of field dependency :

1.Now create Field Dependency For District field,Open Object Manager,Choose Farmer object ,select Fields&Relationships,On Right side We have Field Dependency option,select that option.

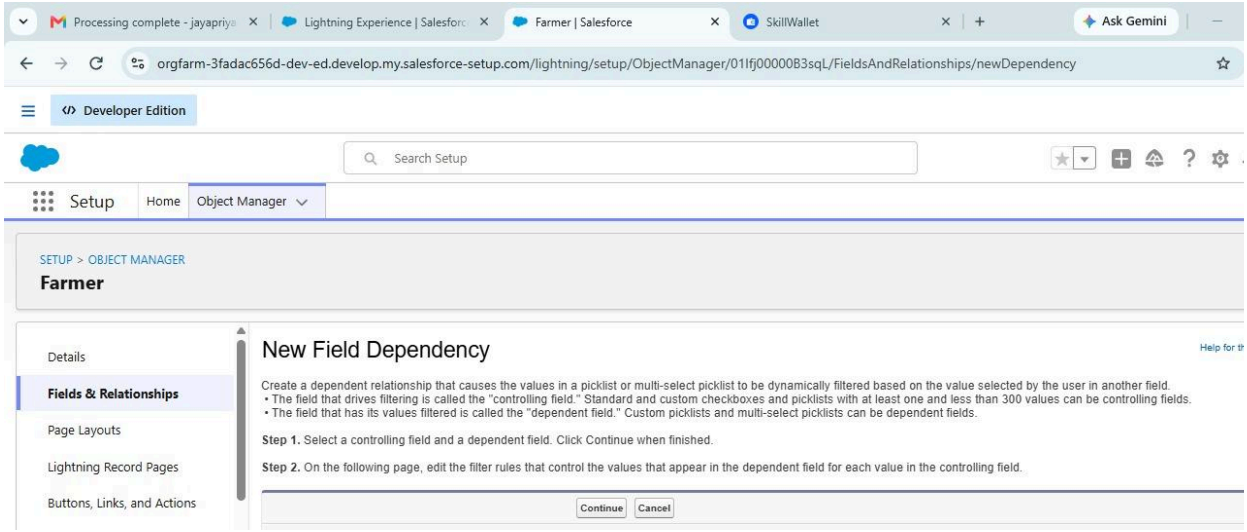
2.Click new, Select Controlling Field as State,Dependent Field As District,Click on Continue.

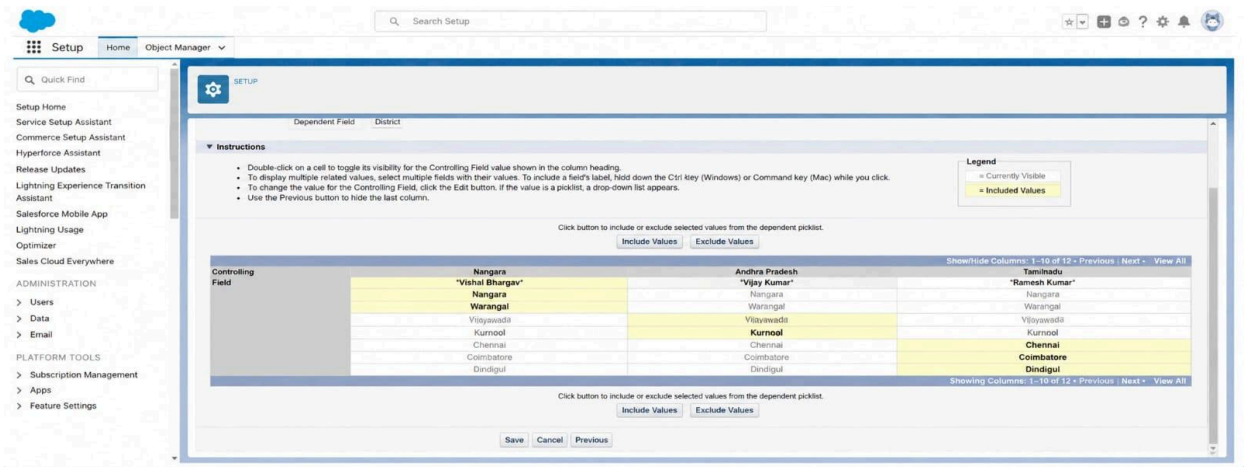
3.Select the appropriate Value in each column by double-clicking them.

4. For Ex. Telangana - Yadadri Bhongir.

5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.

6. Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.





7.Click Close to close the preview window, Click Save.

Activity 4:

Creation of fields for the Buyer objects

- **Buyer:** Contact information and Buyer Type categories.

1.Select Data type as a Phone,Enter a Field Label(Contact),Save.

2.Select Data type as an Email,Enter Field Label(Email), Next save.

3.Select Data type Text area,Enter Field Label(Address),Save it.

4.Select Data type Picklist, Field Label(Buyer Type),Enter values (Wholesale,Retail),Click Save.

Milestone 6 : Profile

What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do in
salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab
settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

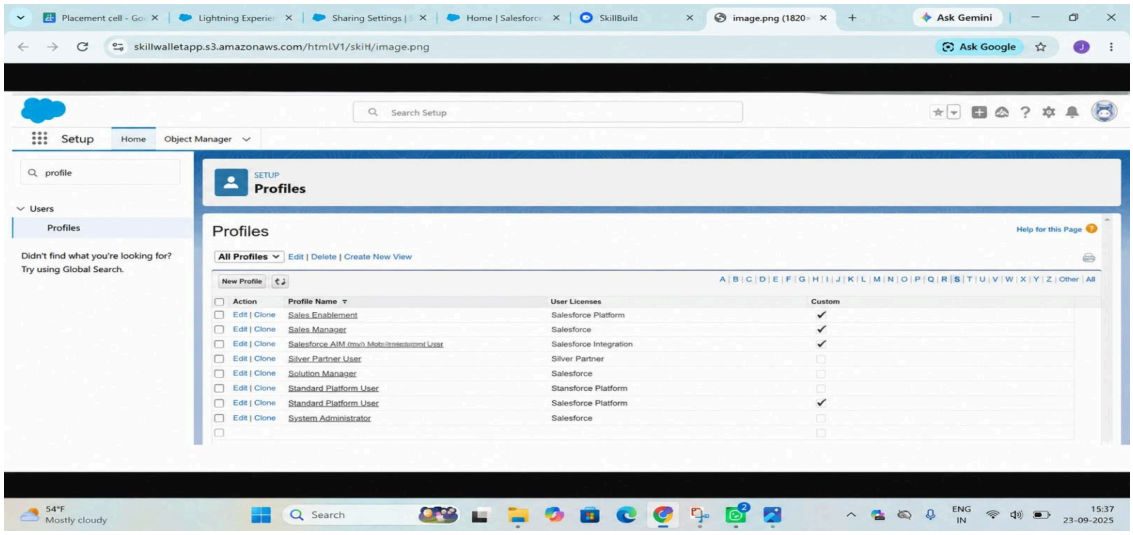
A profile can be assigned to many users, but users can be assigned a single profile at a time.

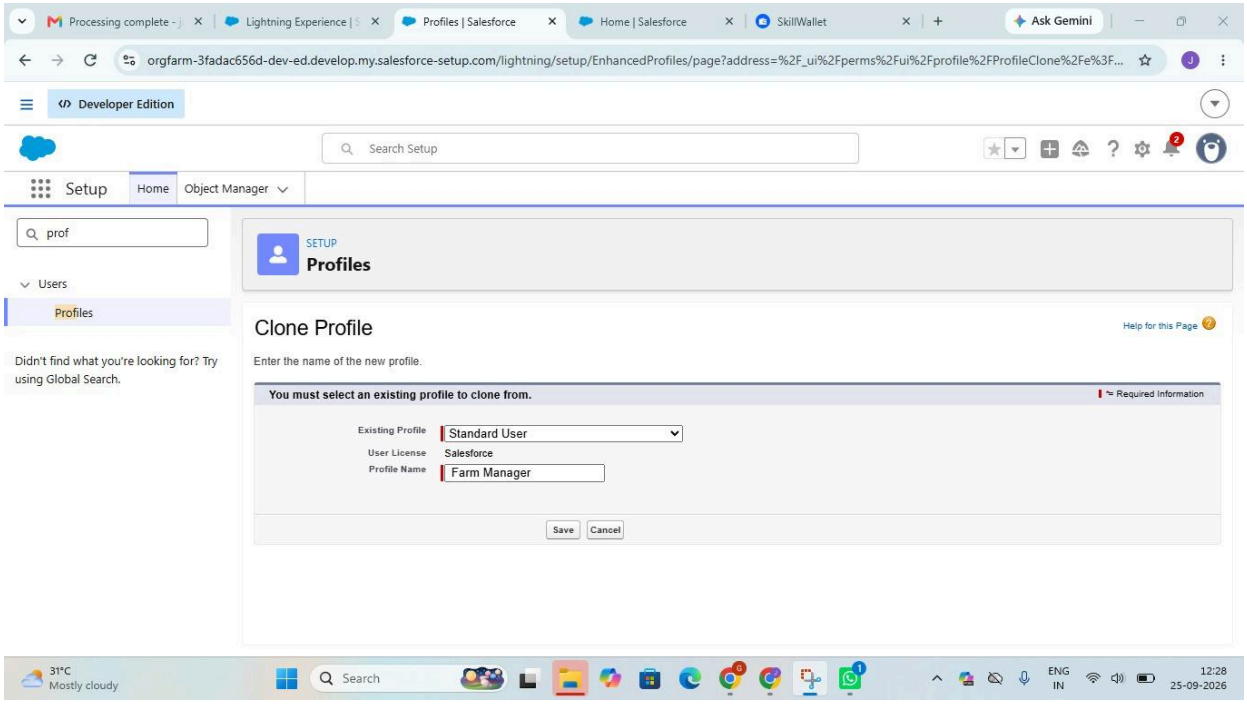
Activity 1:

Create a custom profile

- **Profiles:** Created "Sales Department" and "Farm Manager" clones of the Standard User.

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter the Sales Department.





Activity 2:

Create a custom profile-2

Create a profile with the profile name as “ Farm Manager” .

6. From setup , enter profiles in Quick Find box

7. Select profiles (Standard user).

8. Click clone.

Milestone 7: Role

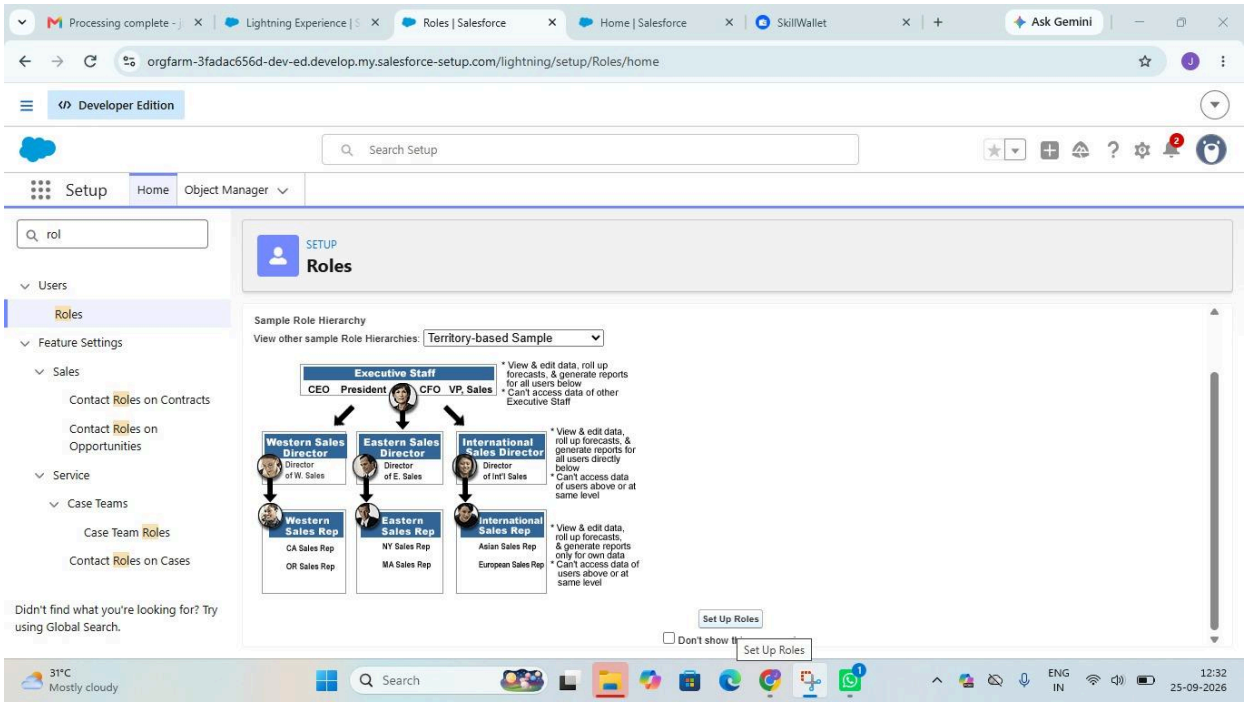
In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

Activity-1:

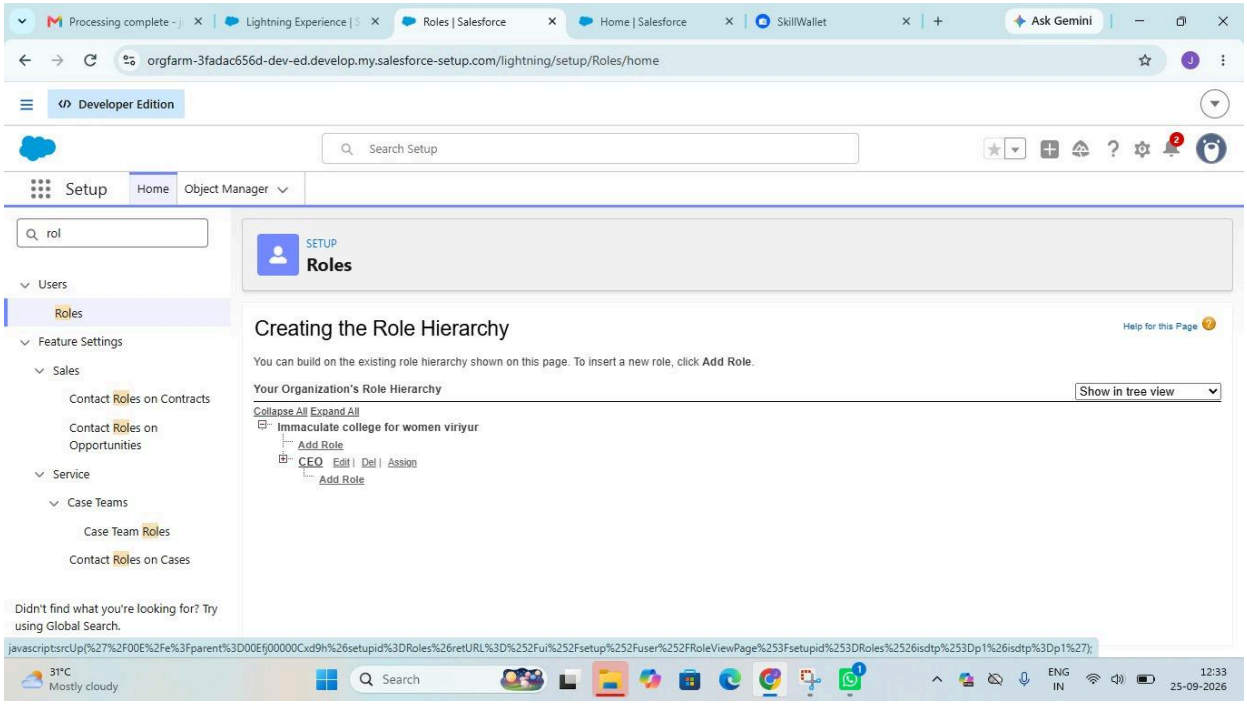
Creation of role

- **Roles:** Implemented a hierarchy with "Farm Manager" reporting to the CEO.

1. From the Quick find box search for the role and click on the roles option
2. select the set up roles option



3.Below the CEO click on add role and enter the label name as a " Farm Manager " and the role name will be Automatically populated and click on save.



Milestone 8: Users

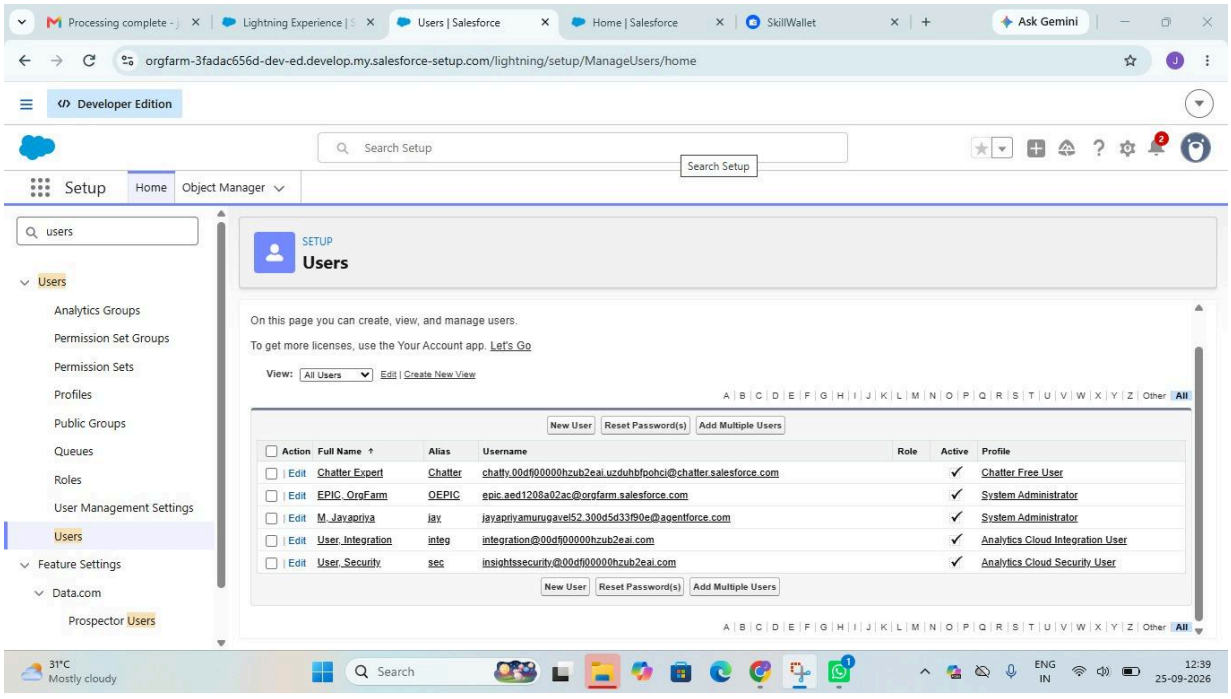
What is a user?

A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in
Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.
NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don't find
salesforce license then deactivate a user who has salesforce license Or change the license type from Salesforce to any other.

Activity 1:

To Create a user

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.



3. Enter First name as Farm and last name as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5. Select role Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile(Farm Manager).
8. Click save.

Activity 2:

Create a user with a username as Sales Department

Create a user with a username as "Sales Department", and assign him the sales Manager profile. Follow the steps from above Activity

Milestone 9: Sharing rules

What are Sharing Rules?

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,

•Owner-based Sharing Rules

•Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

1)Go to Sharing Settings, which can be found under the Quick Find section.

2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.

3) Add the label of the sharing rule you want to make.

4) Select your rule type based on the criteria.

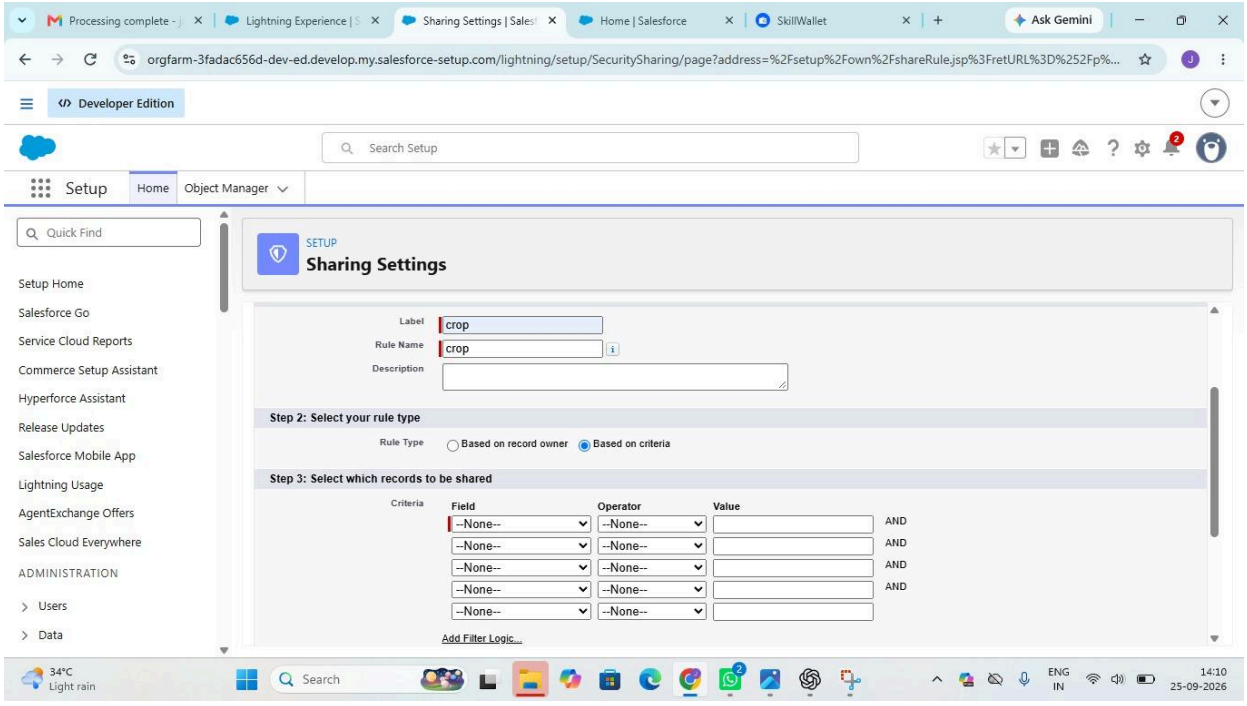
5) Select the field and immediately check the field from the Crop object.

6) Select the crop as equal and the value is True.

7) And in selecting the users to share with the section select roles and in that select Farm Manager.

8) And in the section of select the level of access for the users give the access Read/Write.

9) And save the rule.



Activity 2:

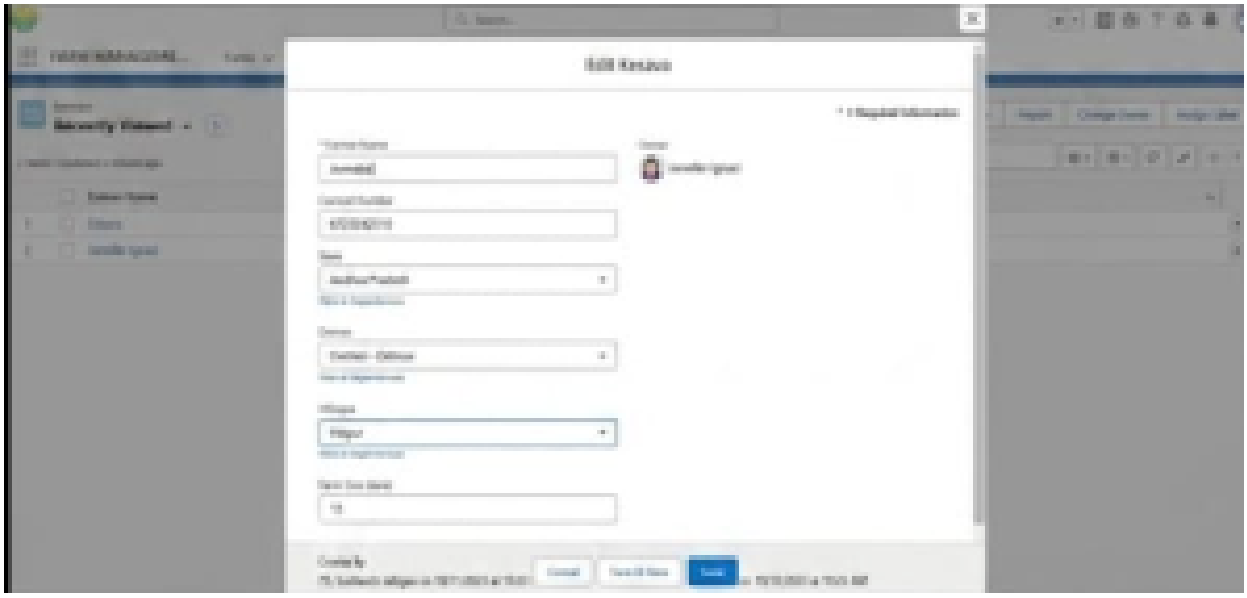
Create a Sharing Rule to Share the records of Buyer

Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.
Follow the steps from above Activity.

Milestone 10: Users Adoption

Activity 1:
Create a Record(Farmer)

- Click on App Launcher on the left side of the screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.



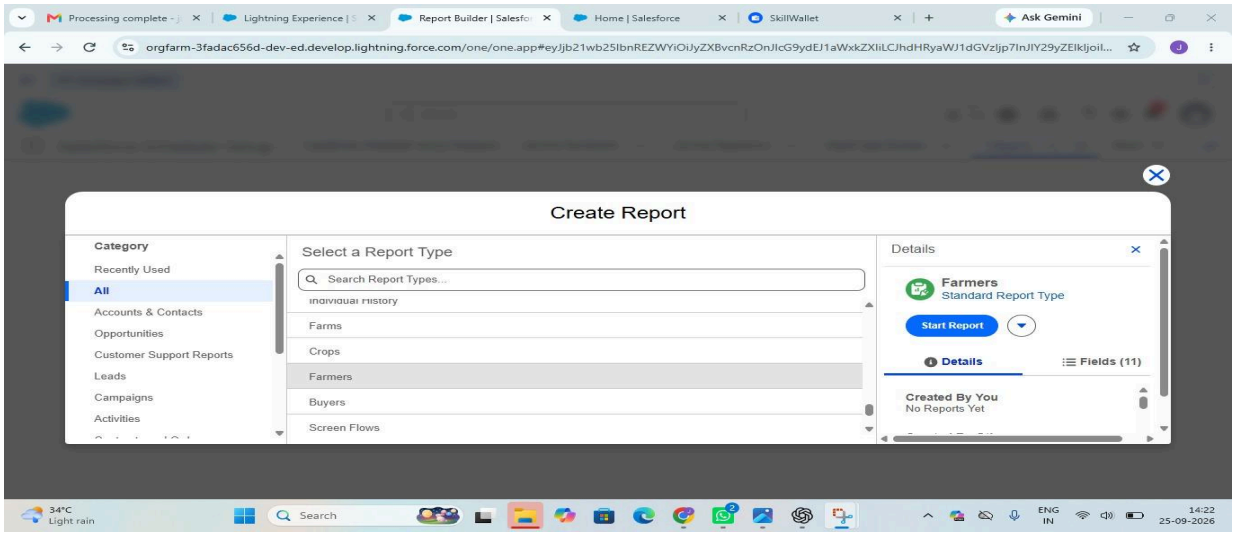
Milestone 11 : Reports

Activity 1:
Create a report

Create a report that displays the rating of the account and which has type and account name.

1) Click on app launcher search for reports.

2) Click on the new report and select the category as Farmer with Farms.



3) In the details section select the option start report.

4) In Outline Click on Column Select Which Field you want to display in the Report.

5) Click Group,Select Which Fields You Want to Group.

6) Save the report by giving the label name(Farmer with Farms) and save the folder as a public folder and save the report.

Activity 2

:Create a Report using the Objects Crops, Buyers.

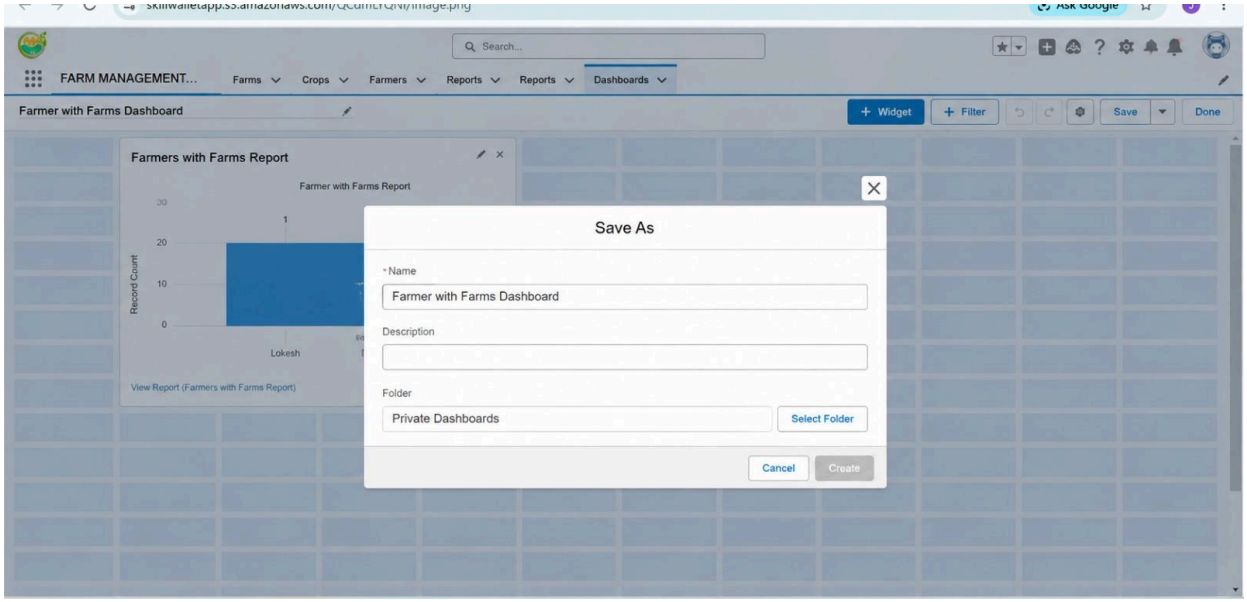
Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

Milestone 12 : Dashboard

Activity 1 :

Create a Dashboard

1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.



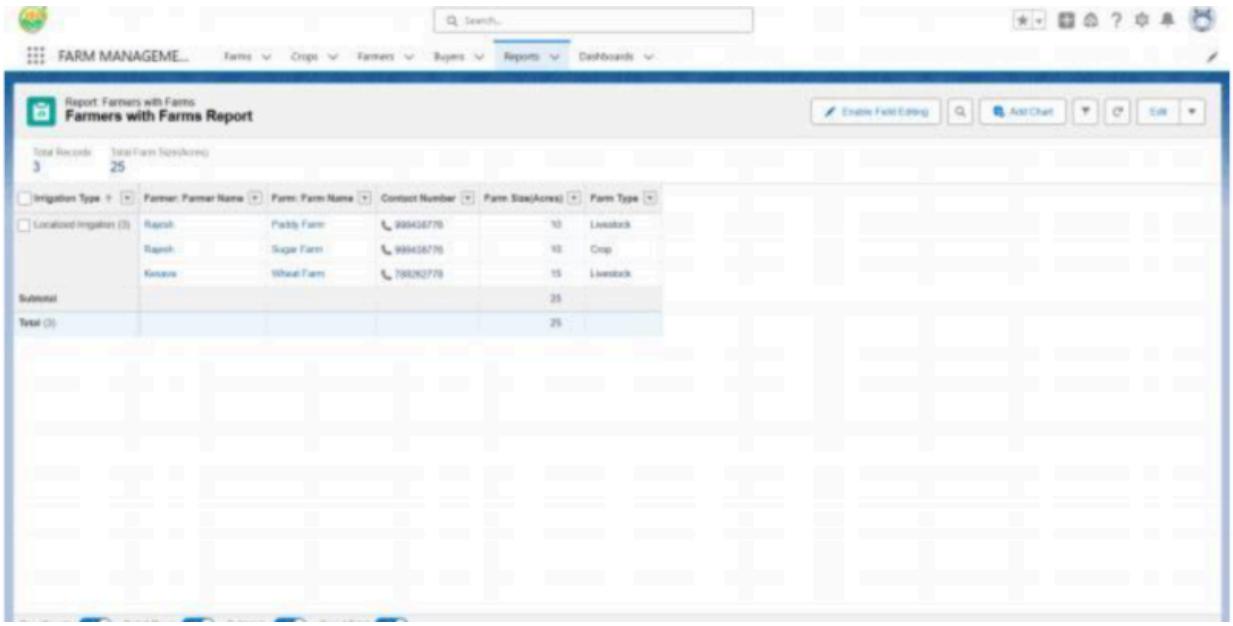
Milestone 13 : View report and Dashboard

Activity 1 :

Report

- Click on App Launcher on the left side of the screen.

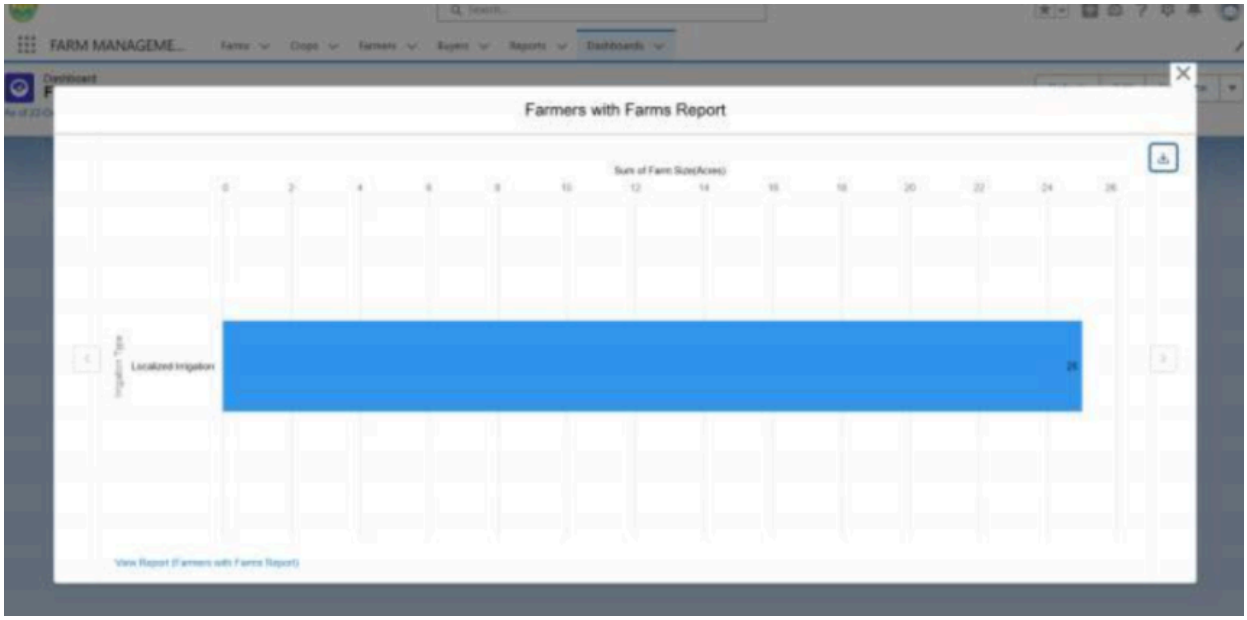
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records



Activity 2:

Dashboard

- Click on App Launcher on the left side of the screen.
- Search Farm Management system & click on it.
- Click on Dashboard Tab.
- Click Farmers With farms & see records.



7. FUNCTIONAL & PERFORMANCE TESTING

The system underwent rigorous testing cycles:

- **UAT:** Verified that Farmers and Managers can perform core tasks.
- **Validation Testing:** Confirmed that harvest dates cannot precede planting dates.
- **Security Testing:** Ensured "Sales Department" users cannot delete "Farm" records.

8. ADVANTAGES & DISADVANTAGES

Advantages:

- Centralized data source.
- Drastic reduction in manual paperwork.
- Real-time visibility into crop health and sales.

- Secure, role-based data access.

Disadvantages:

- Requires stable internet connectivity for real-time updates.
- Salesforce licensing costs for commercial deployment.
- Limited offline data entry capabilities.

9. CONCLUSION

The Farm Management System successfully digitizes agricultural data management using Salesforce. By consolidating fragmented data into a unified cloud platform, the system provides a foundation for more efficient, transparent, and profitable farming operations.

10. FUTURE SCOPE

- **Mobile Integration:** Dedicated Salesforce Mobile App customization.
- **AI/ML:** Einstein Analytics for crop yield prediction.
- **IoT:** Integration with soil moisture and weather sensors.
- **Market Integration:** Live pricing feeds for crop commodities.

11. APPENDIX

- **Tools Used:** Salesforce Developer Edition, Lightning App Builder, Report Builder.
- **Contact:** Person (Lead Developer)